

C 0 CA

Quiz

1100 0000 1100 1010

1 10000₂ 0011001010↑ negative ↑₁₆

$$E = \overset{\text{stored}}{16} - \overset{\text{bias}}{15} = 21$$

normalize mantissa

$$0 \cdot 2^{-1} + 0 \cdot 2^{-2} + 1 \cdot 2^{-3} + 1 \cdot 2^{-4} + 0 \cdot 2^{-5} + 0 \cdot 2^{-6} + 1 \cdot 2^{-7} + 0 \cdot 2^{-8} + 1 \cdot 2^{-9} + 0 \cdot 2^{-10}$$

$$= \frac{1}{8} + \frac{1}{16} + \frac{1}{128} + \frac{1}{512} + \underline{\quad}$$

$$= 1.197265625 \times 2^1 = 2.39453125$$

apply the negative again

- 2.39453125

Exam 1

Powers

1. $10100101_2 \times 10_2$

convert to decimal

128 64 32 16 8 4 2 1
1 0 1 0 0 1 0 1

128
32 > 160
4 > 5
1

$165_{10} \times 2_{10}$
330₁₀

10100101
→ x 10

00000000
101001010
(1)01001010
128 64 32 16 8 4 2 1
64₁₀ + 8₁₀ + 2₁₀ = 74₁₀

2. convert to decimal using two's complement

01010101
flip, then add 1

10101010
+ 1
10101011 = -85

10111000
7 6 5 4 3 2 1 0
-2⁷ + 0⁶ + 2⁵ + 2⁴ + 2³ + 0² + 0¹ + 0⁰
= -128 + 32 + 16 + 8 = -72
-85

1. 8 spaces

0 1 0 1 0 1 0 1
128 64 32 16 8 4 2 1 = 85₁₀

2. 85 - 64 = 21
21 not bigger
21 - 16 = 5
5 - 4 = 1

3. Convert numbers

i.) Convert $(1023)_5$ to base-8 (radix=8)

$$\textcircled{1} \quad \frac{1}{5^3} \frac{0}{5^2} \frac{2}{5^1} \frac{3}{5^0} = (1 \times 5^3) + (0 \times 5^2) + (2 \times 5^1) + (3 \times 5^0) = 138_{10}$$

$$\textcircled{3} \quad \begin{array}{r} 17 \\ 8 \overline{) 138} \text{ R } 2 \\ 8 \overline{) 17} \text{ R } 1 \\ 8 \overline{) 2} \text{ R } 2 \end{array} \quad \uparrow \rightarrow \boxed{212_8}$$

ii.) 762_8 to base-5

$$8^2 \frac{7}{8^1} \frac{6}{8^0} = (7 \times 8^2) + (6 \times 8^1) + (2 \times 8^0) = 448 + 48 + 2 = 498_{10}$$

$$\begin{array}{r} 100 \\ = 500 \\ 5 \overline{) 498} \text{ R } 3 \\ 5 \overline{) 19} \text{ R } 4 \\ 5 \overline{) 3} \text{ R } 4 \\ 5 \overline{) 3} \text{ R } 3 \end{array} \quad \uparrow \quad 3443_5$$

4. Convert unsigned to decimal

$$\begin{array}{r} 0111.1011 \\ \hline 8 \quad 4 \quad 2 \quad 1 \quad \cdot \quad \frac{1}{2} + 0 + \frac{1}{8} + \frac{1}{16} \\ 4+2+1 \\ \hline 7.6875 \end{array}$$

$$\begin{array}{r} 10101.101 \\ \hline 16 \quad 8 \quad 4 \quad 2 \quad 1 \quad \cdot \quad \frac{1}{2} + 0 + \frac{1}{8} \\ 16+4+1 \\ \hline 21.625 \end{array}$$